Git Init

Introduction

Imagine you are starting a **new coding project**. You want to **track all changes**, **collaborate with others**, and **restore previous versions** if needed.

How do you start tracking your project using Git?

This is where the git init command comes in!

Section 1: Learn

What is git init?

git init **is the command that initializes a new Git repository.** It tells Git to start tracking the files in your project.

Why Do We Need git init?

Without Git (git init)	**With Git (git init)**
Changes are **not tracked**	Every change is **saved with history**
No version history	Can **restore previous versions**
Difficult to collaborate	Enables **team collaboration**
No backup of files	Stores **code securely on GitHub**

How Does git init Work?

1. Creates a **hidden** .git **folder** in your project directory.
2. Starts **tracking all changes** in the project.
3. Allows you to **add, commit, and push code** to a remote repository.

An Interesting Anecdote

Before Git, developers used **manual file backups** (project_final.doc, project_final_v2.doc). This caused **confusion and lost work**. With Git, everything is **automated and structured**, thanks to git init.

Section 2: Practice

Step-by-Step Guide to Using git init

1. Installing Git

Before using Git, ensure it is installed.

```
# Check if Git is installed
git --version


# If Git is not installed, install it (Ubuntu)
sudo apt update
sudo apt install git
```

Why is this important?

- Ensures Git is **installed and ready to use**.
- Allows you to **track and manage versions**.

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2. Initializing a Git Repository

```
# Create a new project folder
mkdir my_project
cd my_project

# Initialize Git in the folder
git init
```

What happens here?

- A **hidden** .git **folder** is created inside my_project.
- Git **starts tracking files and changes**.

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3. Checking if Git is Initialized

```
# Check the Git status
git status
```

Expected Output (if no files are added yet):

```
On branch main
No commits yet
nothing to commit (create/copy files and use "git add" to track)
```

This means Git is **ready**, but no files are being tracked yet.

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4. Adding and Committing Files to Git

```
# Create a new file
echo "print('Hello, Git!')" > script.py
```

```
# Stage the file
git add script.py

# Commit the file with a message
git commit -m "Added script.py file"
```

What does this do?

- `git add` **prepares the file for tracking**.
- `git commit` **saves the file with a timestamp and message**.

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5. Checking the `.git` Folder

If you list hidden files in your project, you will see `.git`:

```
ls -la
```

Output:

```
drwxr-xr-x  3 user user  4096 Feb 26 10:00 .git
```

Why is this important?

- The `.git` folder stores **all version control data**.
- If deleted, the repository will **lose its version history**.

Real-World Example

A **startup is building a mobile app**. Their **team of 5 developers** needs to:

1. **Track every change in the project.**
2. **Restore previous versions if needed.**
3. **Collaborate without overwriting each other's work.**

What did they do?

1. Ran git init in their project folder.
2. Used git add **and** git commit to track their changes.
3. Pushed their repository to **GitHub** for remote collaboration.

Result? Faster development, **better teamwork, and version control**.

Section 3: Know More

Frequently Asked Questions

1. **What does** git init **do?**

   It **creates a new Git repository** in your project folder, allowing version control.

2. **Can I undo** git init**?**

   Yes! Delete the .git folder:

```
rm -rf .git
```

3. This removes **all Git history and tracking**.
4. **How do I check if a directory is a Git repository?**

```
git status
```

5. If Git is initialized, it will show the **current branch and status**.
6. **Do I need to run** git init **for every project?**

   Yes! Every new project must be initialized separately.

7. **Where can I practice Git commands?**
   - Create a **GitHub account** and start a project.
   - Use **GitHub Desktop** for a graphical interface.

Conclusion

The git init command is **the first step in using Git**. It enables developers to **track, manage, and collaborate on code efficiently**.

Start by **initializing a repository**, add files, and commit changes. Soon, you'll be **managing projects like a pro!**